

One-Step Equations with Rational Coefficients

Answer Key

Grade 6–7 Mathematics

Quick Answers

- | | | |
|-------------------|------------------|-----------------|
| 1. 5 | 5. 25 | 9. -1.2 |
| 2. 20 | 6. 12 | 10. (b) = 45, — |
| 3. $\frac{11}{4}$ | 7. 15 | |
| 4. $\frac{7}{2}$ | 8. $\frac{6}{5}$ | |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6–7 Mathematics

7.EE.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 11 tagged checks.

- 1.** Solve (balance model): $x + 7 = 12$

Solution: Undo the $+7$ by subtracting 7 from both sides (removing 7 unit weights from both pans): $x = 12 - 7$. Check: $5 + 7 = 12$. ✓

$x = 5$

- 2.** Solve (tape model): $\frac{x}{5} = 4$

Solution: The strip x is 5 equal parts of 4 each, so undo the division by multiplying both sides by 5: $x = 4 \cdot 5$. Check: $20 \div 5 = 4$. ✓

$x = 20$

- 3.** Solve: $x - \frac{3}{4} = 2$

Solution: Undo the subtraction by adding $\frac{3}{4}$ to both sides: $x = 2 + \frac{3}{4}$. Check: $\frac{11}{4} - \frac{3}{4} = \frac{8}{4} = 2$. ✓

$x = \frac{11}{4}$

- 4.** Solve, writing the answer as a fraction: $6x = 21$

Solution: Undo the multiplication by dividing both sides by 6: $x = \frac{21}{6}$, which simplifies by 3 to $\frac{7}{2}$. Check: $6 \cdot \frac{7}{2} = 21$. ✓

$x = \frac{7}{2}$

- 5.** Solve: $0.4x = 10$

Solution: Divide both sides by 0.4: $x = 10 \div 0.4 = 25$. Check: $0.4 \cdot 25 = 10$. ✓

$$x = 25$$

6. Solve (bar model): $\frac{2}{3}x = 8$

Solution: Multiplying by the reciprocal of $\frac{2}{3}$ undoes it in one step: $x = 8 \cdot \frac{3}{2} = 12$. (Model view: 2 parts total 8, so each part is 4, and the 3 parts of the whole bar total 12.)

Check: $\frac{2}{3} \cdot 12 = 8$. ✓

$$x = 12$$

7. Solve: $\frac{x}{2.5} = 6$

Solution: Multiply both sides by 2.5: $x = 6 \cdot 2.5 = 15$. Check: $15 \div 2.5 = 6$. ✓

$$x = 15$$

8. Solve, writing the answer as a fraction in simplest form: $\frac{3}{4}x = \frac{9}{10}$

Solution: Multiply both sides by the reciprocal $\frac{4}{3}$:

$$x = \frac{9}{10} \cdot \frac{4}{3} = \frac{36}{30} = \frac{6}{5}.$$

Check: $\frac{3}{4} \cdot \frac{6}{5} = \frac{18}{20} = \frac{9}{10}$. ✓

$$x = \frac{6}{5}$$

9. Solve: $x + 2.6 = 1.4$

Solution: Subtract 2.6 from both sides: $x = 1.4 - 2.6$. Subtracting a larger number from a smaller one gives a negative result. Check: $-1.2 + 2.6 = 1.4$. ✓

$$x = -1.2$$

10. Challenge. Two-fifths of a number x is 18. (a) write the equation; (b) solve it.

Solution: (a) “Two-fifths of the number” means multiplication, so the sentence in symbols is $\frac{2}{5}x = 18$ (also fine: $2x/5 = 18$ or $0.4x = 18$ — instructor judges the equation).

(b) Multiply both sides by the reciprocal $\frac{5}{2}$: $x = 18 \cdot \frac{5}{2} = 45$. Check: $\frac{2}{5} \cdot 45 = 18$. ✓

$$x = 45$$